

Mediguide: Reducing Cognitive Load and Improving Elderly Patients' Healthcare Experience

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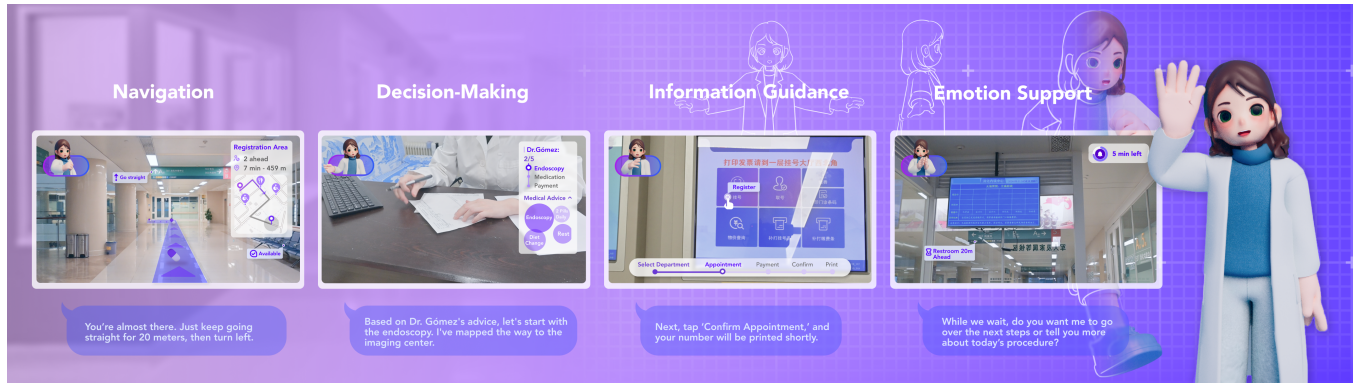


Figure 1: Seattle Mariners at Spring Training, 2010.

Abstract

Older adults often experience psychological stress and physical discomfort during medical visits, as they must simultaneously manage various tasks such as wayfinding, making payments, and understanding medical results. Traditional tools like signage, mobile apps, and asking for help often fall short due to unclear information and complex interfaces, increasing cognitive load and potentially diminishing their overall healthcare experience.

To address these challenges, we developed Mediguide, a virtual assistant that provides personalized navigation support through multimodal interactions and dynamic adjustments. Mediguide aims to help older adults receive clear, accurate information, supporting them in navigation and decision-making processes. A comparative study using a virtual reality (VR) hospital simulation demonstrated reduction in cognitive load, and a decrease in decision errors, and significant improvements in both autonomy and overall healthcare experience.

Mediguide supports older adults in managing complex tasks during medical visits and offers practical insights for designing age-inclusive healthcare systems.

CCS Concepts

• **Human-centered computing** → **Usability testing**; **Empirical studies in HCI**; **Empirical studies in interaction design**.

Keywords

Hospital user experience, Older Adults, Virtual Agent

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1 Introduction

As the global population ages, older adults increasingly face challenges during medical visits, particularly in complex healthcare facilities. They not only experience physical discomfort but also face significant cognitive demands while managing tasks such as wayfinding, making payments, booking appointments, and understanding medical instructions. [5, 13, 19, 25, 26]. In many cases, older adults need to manage these tasks independently, often with limited support [11], which adds to the complexity of navigating healthcare facilities.

Healthcare facilities in China, like many other regions, typically provide multiple digital systems to assist patients during medical visits, including mobile apps, mini-programs, payment systems, and medical record management platforms. However, these tools are often fragmented, requiring users to switch between different systems for various tasks. Moreover, these systems are typically not designed with older adults in mind, leading to poor usability

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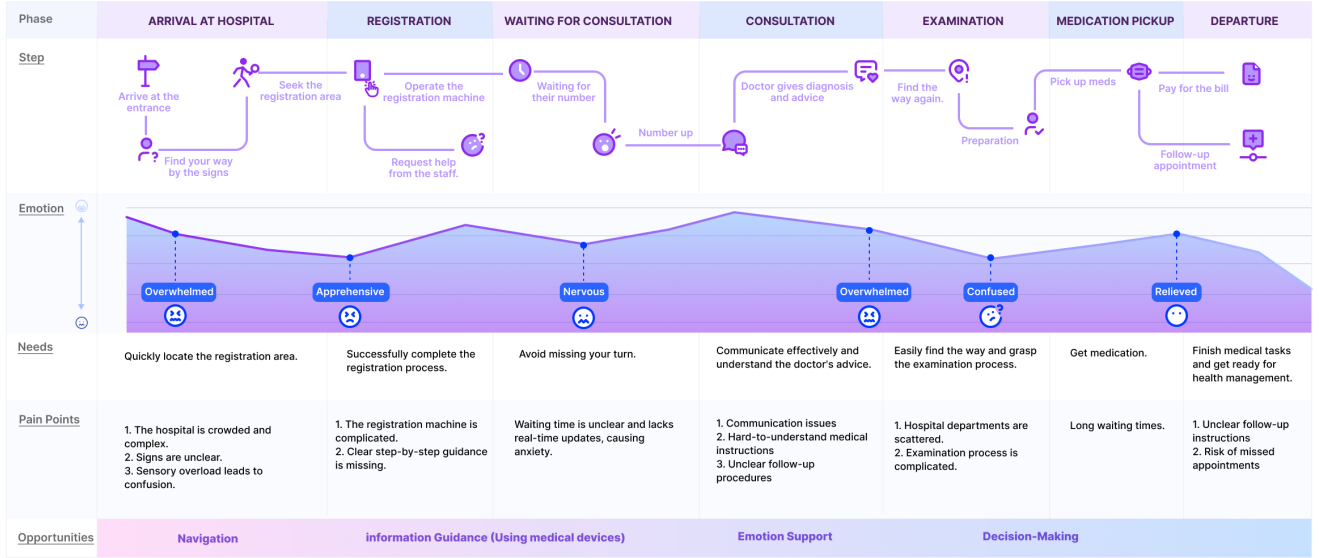


Figure 2: Older adults' journey map in hospital

[7]. For example, mobile apps and mini-programs tend to have complex interfaces that create cognitive barriers for older adults. [12, 21, 25], while signage systems often face accessibility limitations with poor contrast ratios, insufficient text size, and outdated design principles. [4, 18]. In-person assistance can be challenging due to communication barriers, particularly for older adults with hearing impairments or difficulties comprehending verbal instructions. This communication challenge, combined with other fragmented and poorly integrated support systems in healthcare facilities, contributes to increased cognitive demands, information processing difficulties, and potential decision-making errors during medical visits [15, 24].

To address these challenges, we used a virtual assistant, which demonstrated significant potential for enhancing the healthcare experience of older adults [1, 6, 9, 20]. These assistants can provide context-aware, personalized support, helping older adults with wayfinding, information processing, and decision-making tasks. This research introduces Mediguide, a virtual assistant designed to provide comprehensive support to older adults throughout their healthcare journey. Building upon established research in virtual assistance and user-centered design principles for aging populations, Mediguide implements visual and auditory feedback with dynamic adaptations based on real-time user needs. In contrast to existing static support systems, Mediguide delivers personalized guidance to ensure older adults receive clear, concise, and contextually relevant information aligned with their specific tasks, thereby reducing navigation errors and cognitive burden in decision-making processes. The system's naturalistic interface promotes trust and self-efficacy, enabling older users to navigate complex healthcare procedures with increased confidence and reduced anxiety.



Figure 3: virtual assistant Character and UI Design

To evaluate Mediguide's effectiveness, we conducted a comparative experiment using a virtual reality (VR) simulation of a hospital environment. This study assesses how Mediguide impacts cognitive load, patient autonomy, and the overall healthcare experience for older adults. The results provide insights into how Mediguide helps reduce cognitive overload, minimize decision errors, and improve older adults' independence and satisfaction during medical visits. These findings also offer valuable lessons for designing elderly-friendly healthcare support systems.

2 Mediguide design process

2.1 Virtual Assistant Character Design

Mediguide is a virtual assistant system designed to alleviate anxiety and enhance the overall medical experience for older adults through emotional support and practical assistance. The system's design was informed by interviews with five older adults, who identified common challenges they face in hospital settings, particularly in navigating the environment, understanding medical information,

and making decisions. These challenges were especially evident when patients used self-service machines or accessed digital results. One participant explained, "I always struggle with booking an appointment on APP or equipment in hospital; it seems too complicated." Another patient noted, "The hospital volunteers are helpful I always ask them for wayfinding but they're often too busy. Sometimes I feel more comfortable asking them for help than my own family because they tend to be more patient." "What's more during the Covid-19 period I had to go hospital by myself which makes me so anxious and I really need someone help me even though just tell me don't worry." These insights were pivotal in shaping Mediguide, which aim to reduce cognitive load by providing crucial support in navigation, emotional support, and decision-making, ultimately making the hospital experience more manageable and less stressful for older adults.

To address these needs, we designed Mediguide as a young, friendly female figure characterized by gentle facial expressions, a calm tone, and warm body language to foster trust and provide comfort. Her youthful appearance conveys vitality and hope, which helps mitigate the anxiety frequently experienced by older adults [17, 22, 23]. Our discussions with patients revealed that they value a balance between professionalism and approachability in the assistant's appearance. As one participant noted, "If the virtual assistant looks like a doctor or nurse, I'd feel more confident in following its guidance." Based on this feedback and literature on virtual agents design [16], we decided to portray Mediguide wearing a white lab coat, reinforcing her credibility and professionalism while still maintaining a welcoming and reassuring demeanor. Although designed to be approachable, we intentionally avoided making her overly cute or informal, as this could potentially diminish her perceived authority and reliability.

The color scheme was also carefully chosen to ensure both clarity and usability [8]. A blue and purple palette was selected for its high visibility and distinctiveness, particularly in the visually crowded hospital environment. This color choice aligns with the AR interface, creating a cohesive and seamless user experience. Mediguide is positioned in the upper left corner of the AR display to avoid obstructing the user's view, while still offering clear navigational cues and relevant information. This unobtrusive placement ensures that users can access guidance without distracting them from their immediate surroundings.

Functionally, Mediguide is designed to address the primary challenges older adults face during hospital visits. It provides multi-dimensional support, including navigation assistance, information interpretation, emotional comfort, and decision-making guidance. By utilizing a combination of voice prompts and visual cues, Mediguide delivers clear, precise information, helping older adults feel more confident and empowered throughout their medical journey. This not only reduces cognitive load but also promotes greater independence and peace of mind as patients engage with Mediguide throughout their hospital experience.

2.2 Virtual Assistant States

The virtual assistant transitions between multiple states based on user needs, ensuring a seamless and personalized experience within the hospital environment:

Greeting State: Upon activation, the virtual assistant greets the user warmly and briefly introduces its role as a healthcare guide. Users can ask questions at any time, fostering a welcoming atmosphere and helping reduce anxiety [2, 3]. This state also includes basic audio interaction to gather preliminary information about the user's health status and medical needs, facilitating support for subsequent healthcare decisions.

Idle State: When immediate assistance is not required, the virtual assistant enters an idle state, remaining available without interrupting the user's experience.

Guidance State: During navigation or task guidance, the virtual assistant uses augmented reality (AR) visual cues and gestures to indicate directions and procedural steps. This state is essential for assisting users in navigating the hospital efficiently, reinforcing instructions through both visual and verbal cues.

Communication State: The virtual assistant remains accessible for interaction at any time, allowing users to ask questions or seek additional assistance. Through natural voice interaction, it updates in real-time according to the user's evolving needs, providing continuous support throughout the medical process.

Motivating State: During waiting periods or when inactivity or signs of anxiety are detected, the virtual assistant proactively offers psychological support through polite encouraging words and reassuring gestures, aiming to reduce stress and increase the user's sense of security in the hospital setting.

2.3 Interface

The interface is designed to assist elderly users throughout their medical journey in the hospital, providing comprehensive support at key moments.

Navigation Scenario In this scenario, the virtual assistant guides users through the hospital. These include arrows and path markers projected onto the floor, positioned low in the user's visual field to avoid obstruction. Along the route, floating 3D icons related to the destination are suspended above the path, providing a continuous visual reference for users. A pop-up panel on the right side of the interface offers an overview of the route, displaying the distance to the destination and estimated time of arrival. Important locations, such as food services and accessibility facilities, are marked with icons to help users plan their journey effectively.

Decision-Making Scenario Following consultations with healthcare providers, the virtual assistant offers decision-making support. The doctor's recommendations are displayed as speech bubbles on the interface, with key information highlighted using larger font sizes and distinct colors, making it easier for users to quickly understand critical points. Based on these recommendations, Mediguide suggests next steps in the treatment process and maps out the necessary routes to relevant departments or services.

Information Guidance Scenario When users need to operate hospital equipment, the virtual assistant provides clear, step-by-step instructions through blinking markers. A progress bar at the bottom of the interface updates in real-time, showing the current stage of the process and remaining steps. This guidance minimizes errors and ensures smooth task completion.



Figure 4: The environment setting with and without Mediguide

Emotion Support Scenario During waiting periods, the virtual assistant helps alleviate anxiety by engaging with the user. A countdown timer is displayed in the upper-right corner of the interface to keep users informed about their position in the queue, ensuring they don't miss their upcoming examination or appointment. The virtual assistant monitors the user's emotional state and offers psychological support if signs of restlessness or prolonged waiting are detected. Additionally, the assistant reminds users to prepare for upcoming tests or treatments, ensuring they are both mentally and physically ready for the next stage of their medical journey.

3 Methodology

3.1 Participants

The usability study involved 20 elderly participants aged 60 and above. We divided them into 2 groups. Participants are required to be free from cognitive impairments and can follow basic instructions to finish task. They should have experience on interacting with digital tools, such as being able to use mobilephone for daily tasks. While prior experience with VR products is not mandatory, and if they did they would be preferred. Our experiment has received ethical approval from Tsinghua University. Before the experiment, participants will be asked to sign an informed consent form and pay them 60 RMB per hour of the tasks.

3.2 Tasks

Two groups Participants will complete four key tasks in different scenarios. One group is in virtual reality (VR) simulation of a hospital environment based on the real hospital, and the other in a real environment.

Navigation: Participants were tasked with finding a specific department within the hospital using the guidance provided by the virtual assistant, Mediguide. This task simulated the real-world challenge of navigating complex hospital layouts and tested the assistant's ability to provide clear, actionable directions.

Appointment Management: Participants were required to interact with hospital registration systems to complete a check-in and schedule a follow-up appointment. This task assessed how well Mediguide supported elderly users in managing administrative tasks, which often involve multiple steps and unfamiliar digital interfaces.

Medical Information Understanding: Participants were asked to follow a sequence of steps, such as paying for an X-ray and then proceeding to the radiology department. This task evaluated

Mediguide's role in helping older adults interpret and act on complex medical instructions, a critical aspect of reducing decision-making errors.

Emotional Support During Waiting: During the waiting period, participants interacted with the virtual assistant, which provided emotional support through verbal encouragement and status updates. This task reflected a common experience in hospitals, where long waiting times can increase anxiety, and assessed the virtual assistant's effectiveness in reducing stress and maintaining patient comfort.

These tasks were designed to reflect real challenges faced by older adults in hospital settings and were used to measure the virtual assistant's impact on task performance, error reduction, cognitive load, and emotional support.

3.3 Procedure

In real environment: Participants were asked to finish the task in real environment, we recorded completion times, error rates, and cognitive load. And between each task they need to finish NASA-TLX scale. After that, participants need to take part in semi-structured interviews.

In VR environment Participants first underwent a short training session to familiarize themselves with the VR equipment and the virtual assistant interface. The usability test was then conducted within the VR simulation, where participants interacted with Mediguide to complete the tasks. Observations were made throughout the session, and data on task completion times, error rates, and cognitive load were collected.

After completing the tasks, participants took part in semi-structured interviews to provide feedback on Mediguide's usability.

3.4 Data Collection and Analysis

Quantitative Data: Data on task completion times, error rates, and cognitive load were recorded during each task. Cognitive load was measured using the NASA-TLX scale, which captures participants' perceived workload across different dimensions (e.g., mental demand, effort, and frustration).

Qualitative Data: Semi-structured interviews were conducted to gather in-depth insights into participants' experiences with Mediguide, focusing on ease of use, emotional support, and the perceived effectiveness of the virtual assistant.

4 Results and Findings

The usability study provided both quantitative and qualitative insights into the performance of Mediguide in enhancing older adults' hospital experience. The results are organized based on the four key tasks: navigation, appointment management, medical information understanding, and emotional support during waiting. The following sections detail the performance outcomes, cognitive load reduction, error rates, decision-making consistency, and user satisfaction.

4.1 Navigation Task

Participants using Mediguide completed the navigation task significantly faster than those without the virtual assistant. The average task completion time for the Mediguide group was 5.2 minutes

(SD = 1.1), compared to the control group's 7.8 minutes (SD = 1.4), showing a marked improvement in navigation efficiency, but several limitations encountered in real-world need to be discussed. For instance, the high-density pedestrian traffic commonly found in hospital and the differences between physical walking and VR-based interactions present notable challenges.

In terms of cognitive load, the Mediguide group reported a lower NASA-TLX score of 45.7 (SD = 6.3), while the control group scored 62.8 (SD = 7.9), indicating reduced mental effort when using the virtual assistant.

The error rate for the navigation task was also lower for the Mediguide group, with participants making an average of 1.2 errors (SD = 0.5) compared to 3.4 errors (SD = 0.8) in the control group. This reduction in errors highlights Mediguide's effectiveness in providing clear, actionable directions. Combined with the decision-making consistency, these findings demonstrate that Mediguide may improve navigation accuracy and reduce wayfinding uncertainty in complex hospital environments.

4.2 Appointment Management Task

For the appointment management task, participants were required to complete a hospital registration and schedule a follow-up visit. The Mediguide group completed the task in 6.1 minutes (SD = 1.3), and control group required 5.4 minutes (SD = 1.5). Due to input limitations in VR, certain contents must be input by using controllers, while other interactions can be accomplished through audio communication. We hypothesize that's why time increased.

Participants in the Mediguide group also experienced lower cognitive load, with a NASA-TLX score of 43.2 (SD = 5.9), compared to 60.5 (SD = 6.7) for the control group.

Regarding error rates, Mediguide users made fewer mistakes, with an average of 0.8 errors (SD = 0.3), while the control group had an error rate of 2.1 (SD = 0.6). This suggests that Mediguide significantly reduced confusion and improved task accuracy during the administrative process.

4.3 Medical Information Understanding Task

In the medical information understanding task, participants had to follow a sequence of steps, such as paying for an X-ray and proceeding to the radiology department. The Mediguide group successfully completed the task with an average time of 7.4 minutes (SD = 1.2), compared to 9.3 minutes (SD = 1.6) in the control group. Cognitive load in this task was lower for Mediguide users, with a NASA-TLX score of 40.3 (SD = 5.8), compared to 58.5 (SD = 8.1) for the control group.

4.4 Emotional Support During Waiting Task

During the waiting task, participants interacted with Mediguide, which provided verbal encouragement and status updates. Although task completion time was not applicable in this scenario, participants in the Mediguide group reported it can lower anxiety levels, as captured through qualitative interviews and feedback. Participants noted that the virtual assistant's presence made them feel more at ease during long waiting periods, they mentioned: "Having the assistant there made me feel less stressed while waiting. It was reassuring to know that someone was keeping me updated."

—Mr.Wang "You know what? Besides encouragement talk, I found the navigation can super reassuring too. It was my first time at this hospital, and I was pretty nervous, I don't know if it can work or not. But with the spot-on directions from the agent, I didn't need to know every little detail about where am I – I could still find my way. That really took a load off my mind.—Mrs.Li

However, there still have different voice: "Now, since this wasn't a real doctor's visit, I couldn't help but compare it to my own experiences. I'm not sure if this virtual assistant would be as helpful if I was actually sick and at the hospital for real. But for now, at least, I'd say it's not too shabby."— Mr.Ma

5 Discussion

Across all four tasks, Mediguide demonstrated improvements in task completion time, cognitive load reduction, and error rates, as well as enhanced emotional support and decision-making consistency. However, the study also revealed some noteworthy details about how older adults interacted with Mediguide, as well as areas that require further consideration

Many elderly participants were initially hesitant to interact with Mediguide, especially when using the virtual assistant for more complex tasks such as medical information understanding. Some participants expressed concerns about trusting a digital system to handle important medical decisions. For instance, one participant remarked, "I'm not sure if I can rely on it completely when it comes to understanding what the doctor said." However, as participants became more familiar with the assistant, their trust gradually increased, especially in tasks like navigation and emotional support.

Despite the positive reception of Mediguide, some elderly users expressed concerns about the assistant's decision-making suggestions. Several participants preferred to double-check the virtual assistant's recommendations with researchers. This raises important questions about how elderly users perceive the reliability of virtual systems, especially in high-stakes scenarios, and it suggests the need for a hybrid approach where the virtual assistant complements rather than replaces human support.

While, compared with other research, Lee J D and See K A [10] propose that transparency, predictability, and demonstrated competence are fundamental to establishing trust in virtual systems. This framework helps explain why Mediguide users showed increasing confidence as they experienced consistent and reliable interactions, particularly in lower-stakes tasks like navigation.

In healthcare settings specifically, Bickmore et al. [3] have demonstrated that virtual nursing agents can establish lasting trust relationships through empathetic interactions and clear capability demonstration. This resonates with our findings where Mediguide's emotional support features received positive feedback, particularly after users had multiple successful interactions with the system. However, as Pak and McLaughlin [14] note in their research on designing for older adults, trust-building with elderly users requires special consideration of their unique needs and concerns.

1.Progressive Capability Demonstration: Start with low-risk tasks and scale complexity with user confidence. 2.Transparent Decision Support: Explain recommendations clearly, show data sources,

present alternatives when uncertain. 3. Complementary Human Integration: Enhance rather than replace human care, enable expert verification and incorporate user feedback.

Participants frequently mentioned that the virtual assistant's visual and auditory cues were clear and easy to follow. However, some elderly users found the interaction pace too fast at times, particularly when receiving multiple pieces of information simultaneously. A few participants suggested that Mediguide should allow for more personalized pacing or offer an option to repeat instructions as needed. This highlights the importance of designing virtual assistants with adaptive features that cater to varying levels of cognitive and physical abilities among elderly users.

While the study successfully demonstrated the effectiveness of Mediguide, it also highlighted areas for improvement. In particular, further refinement is needed in ensuring that Mediguide can handle more complex decision-making scenarios where older adults may feel uncertain or overwhelmed. Additionally, the emotional support functionality, though effective, could be expanded to provide more personalized and context-aware responses, especially during long waiting periods.

6 Conclusion

This study set out to address the challenges faced by older adults in hospital environments, particularly in the areas of navigation, decision-making, and emotional support. Existing navigation tools and hospital systems often fail to meet the specific needs of elderly users, resulting in increased cognitive load, decision-making errors, and heightened anxiety during medical visits. Our goal was to design and evaluate Mediguide, a virtual assistant, to help alleviate these issues by offering personalized guidance, reducing cognitive load, and providing emotional reassurance.

The results of our usability study demonstrated that Mediguide significantly improves older adults' ability to navigate hospitals, complete administrative tasks, and understand medical information. Across the four key tasks—navigation, appointment management, medical information understanding, and emotional support during waiting—Mediguide reduced task completion times, decreased error rates, and lowered cognitive load. Participants also reported high levels of satisfaction, particularly in terms of the emotional support provided by Mediguide during stressful waiting periods.

This research confirms that a virtual assistant like Mediguide can play a crucial role in enhancing the overall healthcare experience for older adults by addressing the gaps in existing systems. The assistant not only reduces the mental effort required to navigate complex hospital environments but also provides much-needed emotional reassurance, improving both task performance and user satisfaction.

While Mediguide has shown considerable promise, it still has limitations like we should consider using 3D map replace 2D map which will benefit for reduce cognitive load, and our small sample size also may have limitation in our findings. In future research we may focus on enhancing the assistant's capabilities in more complex decision-making scenarios and improving its adaptability to individual user needs. Additionally, expanding the system's emotional support functionality to offer more context-aware and personalized responses could further enhance its effectiveness in stressful hospital settings.

In conclusion, Mediguide provides a promising solution to the challenges older adults face in hospitals, offering a pathway to improve patient autonomy, reduce cognitive load, and enhance the overall medical experience. By addressing both practical and emotional needs, this study contributes to the ongoing development of human-centered healthcare technology tailored to an aging population.

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